WILL Part I (addition)

Closed Algebraic System of Relational Orbital Mechanics (R.O.M.)

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Desmos Project

R.O.M.

Remark 0.1. *R.O.M. does not describe how a body moves under forces; it classifies the algebraically allowed relational states of a bound system. It is not equations of motion but algebraically closed system of allowed states within WILL - allowed free will.*

$$\boxed{\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}} \ (z_\kappa = \text{gravitational redshift})$$

$$\boxed{\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}} \ (z_\beta = \text{transverse Doppler shift})$$

Observational Z Inputs

$Z_{sys}(o) = (1 + z_{\kappa o}(o))(1 + z_{\beta o}(o)) = \tau_{Wo}(o)^{-1}$ (product of gravitational red shift and transverse Doppler shift)
$\tau_{Wo}(o) = \kappa_{Xo}(o) \cdot \beta_{Yo}(o) = (Z_{sys}(o))^{-1}$ (product of projectional phase factors on S^1 and S^2 carriers)
$z_\kappa = \frac{1}{\kappa_X} - 1$ (gravitational redshift)
$z_\beta = \frac{1}{\beta_Y} - 1$ (transverse Doppler shift)
$z_{\kappa o} = \frac{1}{\kappa_{Xo}} - 1$ (redshift at phase o)
$z_{\beta o} = \frac{1}{\beta_{Yo}} - 1$ (transverse Doppler shift at phase o)

Global System Parameters (Fixed for the Orbit)

$\kappa = \sqrt{\frac{R_s}{a}} = \sqrt{\frac{\rho}{\rho_{max}}} = \sqrt{\kappa_p^2(1-e)} = \sqrt{4W} = \sqrt{2(\kappa_o(o)^2 - \beta_o(o)^2)} = \sqrt{\frac{1}{2}(3 - \sqrt{1 + 8\tau_{Wo}(O_o)^2})} = \sqrt{1 - (1+z_\kappa)^{-2}}$ (potential projection at semi-major axis)

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$\kappa_X = \cos(\theta_2) = \sqrt{1-\kappa^2}$ (gravitational phase factor)
$\beta = \frac{\kappa}{\sqrt{2}} = \beta_p\sqrt{\frac{1-e}{1+e}} = \sqrt{2W} = \sqrt{(\kappa_o(o)^2 - \beta_o(o)^2)} = \sqrt{\kappa_o^2 - \frac{\kappa_o^2}{2}\cdot(1+(\frac{1}{\delta_o(o)}-1))} = \beta_o(o)\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2\cdot e\cdot\cos(o)}} = \sqrt{1-(1+z_\beta)^{-2}}$ (kinetic projection on semi major axis)
$\beta_Y = \sin(\theta_1) = \sqrt{1-\beta^2}$ (relativistic phase factor)
$\beta_{int} = \frac{\beta}{\sqrt{1-e^2}}$ (interior kinetic projection)
$\tau_W = \kappa_X\beta_Y$ (relational spacetime factor)
$Q = \sqrt{\kappa^2+\beta^2} = \sqrt{\frac{3}{2}}\ \kappa = \sqrt{3}\ \beta$ (total relational shift (magnitude of state difference))
$R_s = \kappa_o(o)^2 r_o(o) = \frac{2Gm_0}{c^2} = \frac{\Delta_{to}(o)}{\tau_o(o)}\kappa^2\beta c = \frac{T_O c}{\pi}(\kappa_o(o)^2 - \beta_o(o)^2)^{\frac{3}{2}} = t_o(o)\kappa_o(o)^2\cdot c = \frac{r_1 r_2}{r_2-r_1}(\beta_1^2-\beta_2^2) = \frac{a}{2}(3-\sqrt{1+8\tau_{Wo}(O_o)^2}) =$
$= \frac{r_o(o)}{2(2a-r_o(o))}(4a-r_o(o)-\sqrt{(4a-r_o(o))^2 - 8a(2a-r_o(o))(1-\tau_{Wo}(o)^2)}) = \frac{T_O}{\pi}(\sqrt{1-(\frac{1}{1+z_\beta})^2})^3 c$
(Schwarzschild radius -system scale)
$a = \frac{R_s}{\kappa^2} = \frac{R_s}{4W} = \frac{\beta_o c}{\omega}\cdot\frac{\sqrt{1-e^2}}{\sqrt{1+e^2+2e\cos(o)}} = \frac{\Delta_{to}(o)}{\tau_o}\beta c$ (semi-major axis)
$m_0 = \frac{\kappa^2 c^2 a}{2G} = 4\pi\rho a^3$ (mass parameter)
$t = \frac{a}{c}$ (temporal scale of the system)
$W = \frac{\beta^2}{2} = \frac{1}{2}(\kappa_o^2 - \beta_o^2) = \frac{1}{4}\kappa_p^2\ (1-e) = \frac{1}{2}(\kappa_o(o)^2 - \frac{\kappa_o(o)^2}{2\delta_o(o)}) = \frac{1}{2}((1-(1+z_{\kappa o}(o))^{-2}) - (1-(1+z_{\beta o}(o))^{-2}))$ (energy invariant - binding energy)
$\Delta\phi = \frac{3\pi}{2}\frac{\kappa_p^4}{\beta_p^2} = \frac{2\pi Q^2}{1-e^2} = 6\pi\beta_{int}^2$ (precession of perihelion per orbit)
$h_W = a\cdot\beta c\cdot e_Y$ (angular momentum)
$\omega = \frac{\beta c}{a}$ (angular frequency)
$T = \frac{2\pi}{\omega}$ (orbital period)

Eccentricity Relations

$e = \frac{1}{\delta} - 1 = 1 - \frac{2\beta_a^2}{\kappa_a^2} = \frac{2\beta_p^2}{\kappa_p^2} - 1$ (eccentricity derived from closure)
$e_Y = \sqrt{1-e^2}$ (eccentricity's orthogonal value)
$e_X = \frac{1+e}{1-e} = \frac{r_a}{r_p} = \frac{\delta_a}{\delta_p} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (shape factor)

Time Integration

$\omega_o(o) = \frac{\beta\cdot c}{a}\cdot\frac{(1+e\cdot\cos(o))^2}{(1-e^2)^{\frac{3}{2}}}$ (angular frequency at phase o)
$\Delta_{to}(o) = \int_0^o \frac{1}{\omega_\theta(\theta)}\,d\theta = \frac{a}{\beta c}\tau_o = \frac{T_O}{2\pi}\tau_o$ (time duration of given phase interval)
$\tau_o = (1-e^2)^{\frac{3}{2}}\cdot(\int_0^o(1+e\cdot\cos(\theta))^{-2}d\theta) = \frac{1}{\sqrt{1-e^2}}\int_0^o(\frac{t_o(\theta)}{t})^2 d\theta$ (Temporal Phase interval)

Perihelion Relations

$r_p = a(1-e) = \frac{R_s}{\kappa_p^2}$ (radius at perihelion)
$\kappa_p = \kappa\sqrt{\frac{1}{1-e}} = Q_p\sqrt{\frac{2}{3+e}}$ (potential at perihelion)
$\kappa_{Xp} = \sqrt{1-\kappa_p^2}$ (potential phase projection at perihelion) $\beta_p = \frac{V_p}{c} = \frac{\kappa_p}{\delta\sqrt{2}} = \sqrt{\kappa_p^2\cdot\frac{1+e}{2}}$ (kinetic at perihelion)

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$\delta_p = \frac{\kappa_p^2}{2\beta_p^2} = \frac{1}{1+e}$ (closure factor at perihelion)
$Q_p = \sqrt{\kappa_p^2 + \beta_p^2}$ (relational shift at perihelion)

Aphelion Relations

$r_a = a(1+e) = \frac{R_s}{\kappa_a^2}$ (radius at aphelion)
$\beta_a = \sqrt{\beta_p^2 e_X^{-2}} = \beta\sqrt{e_X^{-1}}$ (kinetic projection at aphelion)
$\kappa_a = \sqrt{2W + \beta_a^2}$ (potential projection at aphelion)
$\delta_\mathrm{a} = \frac{1}{1-e} = \frac{\kappa_a^2}{2\beta_a^2}$ (closure factor at aphelion)
$Q_a = \sqrt{\kappa_a^2 + \beta_a^2}$ (relational shift at aphelion)

Phase Variables (Depend on o)

$o =$ orbital phase in radians
$r = r_o(o) = a\frac{1-e^2}{1+e\cos o} = \frac{R_s}{\kappa_o^2}$ (radial distance at phase o)
$\kappa_o = \sqrt{\frac{R_s}{r}} = 1 - (1 + z_{\kappa o}(o))^{-2} = \kappa_p\sqrt{\frac{1+e\cos(o)}{1+e}}$ (local potential at phase o)
$\kappa_{Xo} = \sqrt{1-\kappa_o^2} = (z_{\kappa o}(o) + 1)^{-1}$ (gravitational phase factor at phase o)
$\beta_o(o) = \sqrt{\kappa_o^2 - 2W} = \frac{\kappa_o(o)}{\sqrt{2\delta_o(o)}} = \beta\cdot\frac{\sqrt{1+e^2+2\cdot e\cdot\cos(o)}}{\sqrt{1-e^2}} = \sqrt{\frac{\kappa_o(o)^2}{2}\frac{1+e^2+2\cdot e\cdot\cos(o)}{1+e\cdot\cos(o)}} = \sqrt{\frac{R_s}{r_o(o)} - \frac{R_s}{2a}} = \sqrt{1 - (1 + z_{\beta o}(o))^{-2}}$ (kinetic projection at phase o)
$\beta_R(o) = \beta\ \frac{e\sin(o)}{\sqrt{1-e^2}} = \sqrt{\beta_o(o)^2 - \beta_T(o)^2} = \sqrt{((1-(1+z_{\kappa o}(o))^{-2}) - 2W) - \frac{r_o(o)^2\omega_o(o)^2}{c^2}}$ (radial kinetic projection at phase o)
$\beta_T(o) = \frac{r_o(o)\omega_o(o)}{c} = \beta\frac{1+e\cos(o)}{\sqrt{1-e^2}}$ (transverse kinetic projection at phase o)
$\beta_{Yo} = \sqrt{1-\beta_o^2} = (z_{\beta o}(o) + 1)^{-1}$ (relativistic phase factor at phase o)
$\delta_o = \frac{1+e\cos o}{1+e^2+2e\cos o} = \frac{\kappa_o^2}{2\beta_o(o)^2}$ (local closure factor at phase o)
$Q_o = \sqrt{\kappa_o^2 + \beta_o(o)^2}$ (local relational shift vector at phase o)
$\omega_o = a\beta c\frac{e_Y}{r_o^2} = \frac{\beta c}{a}\frac{(1+e\cos o)^2}{(1-e^2)^{3/2}}$ (angular speed at phase o)
$\eta_o = \frac{r}{a} = 2 - \frac{2\beta_o(o)(o)^2}{\kappa_o(o)^2}$ (phase scale amplitude at phase o)
$\tau_{Wo}(o) = \kappa_{Xo}(o)\cdot\beta_{Yo}(o) = Z_{sys}(o)^{-1}$ (phase spacetime factor at phase o)
$t_o = \frac{r_o(o)}{c}$ (light time scale at phase o)
$\Delta_o = 3\beta_{int}^2 \cdot o$ (precession of perihelion at phase o)

Relational Geometry (WILL)

$\theta_1 = \arccos(\beta)$ (distribution angle on S^1, non-physical)
$\theta_2 = \arcsin(\kappa)$ (distribution angle on S^2, non-physical)
$\Delta_Q = Q_o^2 - Q^2$ (phase relational shift amplitude at phase o)
$O_o = \arccos(1-\delta^{-1}) = \arccos(-e) = \arccos(1 - \frac{2\beta_p^2}{\kappa_p^2}) = \arccos(\frac{2\beta_a^2}{\kappa_a^2} - 1)$ (orbital balance point where $\kappa_o^2 = 2\beta_o^2$ is true)

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Observer dependant

$Z_{raw}(o) = (1 + \beta_{int}(\cos(o + \omega_i) + e\cos(\omega_i))\sin(i))\ Z_{sys}(o)$ (raw light shift including the line of site Doppler at phase o)
$\beta_{los}(o) = \frac{\beta}{\sqrt{1-e^2}}(\cos(o + \omega_i) + e\cos(\omega_i))\sin(i)$ (line of site light shift)
$i =$ (orbital inclination in relation to line of site and orbital plane)
$\omega_i =$ (phase turn or argument of periapsis)
$K_i = \frac{\beta}{\sqrt{1-e^2}}\sin(i) = \beta_{int}\sin(i)$ (semi-amplitude invariant)
$Z_{rawmax} = Z_{sys}(o_{i1})(1 + K_i(1 + e\cos\omega_i))$
$Z_{rawmin} = Z_{sys}(o_{i2})(1 + K_i(-1 + e\cos\omega_i))$
$O_{oi} = (O_o + \omega_i)$
$o_{i1} = -\omega_i$
$o_{i2} = \pi - \omega_i$
$Z_{sysi1}(o_{i1}) = (1 - \beta^2\frac{1+e^2+2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1 - 2\beta^2\frac{1+e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$
$Z_{sysi2}(o_{i2}) = (1 - \beta^2\frac{1+e^2-2e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}(1 - 2\beta^2\frac{1-e\cos(\omega_i)}{1-e^2})^{-\frac{1}{2}}$

Un-tilted 2D coordinates within the orbital plane

$x_{orb} = r(o)\cos(o + \omega_i)$
$y_{orb} = r(o)\sin(o + \omega_i)$

Parametric equations for the observed orbit, entirely free of the absolute ICRF background

$$\begin{bmatrix} x_{sky} \\ y_{sky} \\ z_{depth} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(i) & -\sin(i) \\ 0 & \sin(i) & \cos(i) \end{bmatrix} \begin{bmatrix} x_{orb} \\ y_{orb} \\ 0 \end{bmatrix}$$

$$x_{sky} = r(o)\cos(o + \omega_i)$$
$$y_{sky} = r(o)\sin(o + \omega_i)\cos(i)$$
$$z_{depth} = r(o)\sin(o + \omega_i)\sin(i)$$

Beautiful and Interesting Connections

$\frac{r_a}{r_p} = \frac{1+e}{1-e} = \frac{\beta_p}{\beta_a} = \frac{\kappa_p^2}{\kappa_a^2} = \frac{\kappa_a^2\beta_p^2}{\kappa_p^2\beta_a^2}$ (remarkable equivalence of structural (κ on S^2 carrier) and dynamical (β on S^1 carrier) asymmetry suggesting once again that $SPACETIME \equiv ENERGY$)

1 Operational Independence and the Role of Constants

A frequent objection from conventional frameworks is that any model employing the Schwarzschild radius (R_s) must implicitly depend on the Newtonian constants G and m_0 (mass) as fundamental inputs. This objection assumes that mass is the primary ontological entity and geometry is a secondary derivative.

We demonstrate that this is incorrect. In the WILL RG framework, G and m_0 are not physical inputs but *output calibration tools*. The parameters κ and β are operationally measurable geometric intensities, and R_s is a system scale derived directly from the phase interactions of light.

1.1 Operational Measurability of Relational Projections

Theorem 1.1 (Operational Measurability)**.** *The relational projections are encoded directly in the combined phase interactions of light (spectroscopy) and are operationally independent of G, c, or m_0.*

Proof. **Step 1: The Raw Observable (τ_W).** Spectroscopy does not measure "gravitational potential" or "kinematic velocity" as separate isolates; it measures the total accumulated phase difference between the source and the observer. We define the **Relational Spacetime Factor** τ_W as the inverse of the systems measured redshift product:

$$\boxed{\kappa^2 = 1 - \frac{1}{(1+z_\kappa)^2}} \ (z_\kappa = \text{gravitational redshift})$$

$$\boxed{\beta^2 = 1 - \frac{1}{(1+z_\beta)^2}} \ (z_\beta = \text{transverse Doppler shift})$$

$$\tau_W \equiv \frac{1}{Z_{sys}} = 1/[(1+z_\kappa)(1+z_\beta)]. \tag{1}$$

In the WILL framework, this single observable represents the product of the internal phase projections of the carriers S^2 and S^1:

$$\tau_W = \underbrace{\kappa_X}_{\text{Gravitational Phase}} \cdot \underbrace{\beta_Y}_{\text{Kinematic Phase}} = \sqrt{1-\kappa^2}\sqrt{1-\beta^2}. \tag{2}$$

Step 2: Exact Relational Identity. We reject weak-field approximations. The exact relationship between the signal τ_W and the structural norm Q is given by the algebraic expansion of the phase product:

$$\tau_W^2 = (1-\kappa^2)(1-\beta^2) = 1 - (\kappa^2+\beta^2) + \kappa^2\beta^2. \tag{3}$$

Substituting $Q^2 = \kappa^2 + \beta^2$, we obtain the rigorous link between the optical observable and the geometric state:

$$\tau_W^2 = 1 - Q^2 + \kappa^2\beta^2. \tag{4}$$

This identity demonstrates that the optical signal contains the complete information about the system's structural state, measurable without prior knowledge of mass. □

1.2 Algebraic Determination of System Scale

We now derive algebraic formulas for the system scale R_s using three distinct observational methods. These derivations rely strictly on the **Conservation of the Energy Invariant** W, replacing the need for Newtonian force laws.

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1.2.1 Method A: Differential (Two-Point Method)

This method is suitable when the orbital period is unknown, but the trajectory can be traced geometrically.

Theorem 1.2 (Two-Point Schwarzschild Scale)**.** *Given measurements of geometric position (r) and kinematic intensity (β) at two arbitrary points along a trajectory:*

- *Radii:* r_1, r_2 *(from astrometry)*
- *Intensities:* β_1, β_2 *(from de-projected spectral line widths or proper motion)*

the Schwarzschild radius is:

$$\boxed{R_s = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2).} \tag{5}$$

Proof. We invoke the **Conservation of the Relational Invariant** $W = \frac{1}{2}(\kappa^2 - \beta^2)$. This law states that for any closed system, the specific energy difference between the potential and kinematic projections is constant throughout the orbit.

Therefore, at any two points 1 and 2:

$$\frac{1}{2}(\kappa_1^2 - \beta_1^2) = \frac{1}{2}(\kappa_2^2 - \beta_2^2). \tag{6}$$

Rearranging to group the projections:

$$\kappa_1^2 - \kappa_2^2 = \beta_1^2 - \beta_2^2. \tag{7}$$

Substituting the field identity $\kappa^2 = R_s/r$:

$$\frac{R_s}{r_1} - \frac{R_s}{r_2} = \beta_1^2 - \beta_2^2. \tag{8}$$

Factoring out the scale parameter R_s:

$$R_s(\frac{r_2 - r_1}{r_1 r_2}) = \beta_1^2 - \beta_2^2. \tag{9}$$

Solving for R_s:

$$R_s = \frac{\beta_1^2 - \beta_2^2}{\frac{r_2 - r_1}{r_1 r_2}} = \frac{r_1 r_2}{r_2 - r_1}(\beta_1^2 - \beta_2^2). \tag{10}$$

This formula extracts the "mass" scale purely from geometric gradients. □

1.2.2 Method B: Geometric Resonance (Balance Point Method)

At the specific orbital phase $O_o = \arccos(-e)$, the system passes through its geometric balance point where the instantaneous radius r equals the semi-major axis a. At this unique phase, the closure condition $\kappa^2 = 2\beta^2$ is satisfied instantaneously.

Theorem 1.3 (Balance Point Formula)**.** *Given the geometric scale a and the total light signal τ_W measured at the balance point ($r = a$):*

$$\boxed{R_s = \frac{a}{2}(3 - \sqrt{1 + 8\tau_W^2(O_o)}).} \tag{11}$$

Proof. **Step 1: Spacetime Factor Expansion.** At the balance point ($r = a$), the closure condition implies $\kappa^2 = R_s/a$ and $\beta^2 = R_s/2a$. The observable τ_W is:

$$\tau_W^2 = (1 - \kappa^2)(1 - \beta^2) = (1 - \frac{R_s}{a})(1 - \frac{R_s}{2a}). \tag{12}$$

Step 2: Exact Geometric Solution. Expanding the product:

$$\tau_W^2 = 1 - \frac{R_s}{2a} - \frac{R_s}{a} + \frac{R_s^2}{2a^2} = 1 - \frac{3R_s}{2a} + \frac{R_s^2}{2a^2}. \tag{13}$$

Multiplying by $2a^2$ to clear denominators and rearranging into standard quadratic form ($Ax^2 + Bx + C = 0$):

$$R_s^2 - 3aR_s + 2a^2(1 - \tau_W^2) = 0. \tag{14}$$

Solving for R_s using the quadratic formula ($x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a_{coef}}$):

$$R_s = \frac{3a \pm \sqrt{(3a)^2 - 4(1)(2a^2(1 - \tau_W^2))}}{2}. \tag{15}$$

Simplifying the term under the radical:

$$9a^2 - 8a^2(1 - \tau_W^2) = 9a^2 - 8a^2 + 8a^2\tau_W^2 = a^2(1 + 8\tau_W^2). \tag{16}$$

Thus:

$$R_s = \frac{3a \pm a\sqrt{1 + 8\tau_W^2}}{2} = \frac{a}{2}(3 \pm \sqrt{1 + 8\tau_W^2}). \tag{17}$$

For stable orbits, we must select the negative root. □

Remark 1.4 (Light Separation)**.** *This unit-less factor in brackets is:*

- ***Potential projection at semi-major axis:*** $\kappa^2 = \frac{1}{2}(3 - \sqrt{1 + 8\tau_W^2(O_o)})$

A unique way to separate gravitational part from the light signal.

1.2.3 Method C: Instantaneous (Arbitrary Phase Method)

Most generally, if the orbital geometry (a) is known, the scale R_s can be derived from a **single epoch** observation at any arbitrary radius r_o.

Theorem 1.5 (Arbitrary Phase Formula)**.** *Given the orbital geometry* (a, r_o) *and the single light observable* τ_{Wo}*:*

$$\boxed{R_s = \frac{r_o}{2(2a - r_o)}(4a - r_o - \sqrt{(4a - r_o)^2 - 8a(2a - r_o)(1 - \tau_{Wo}^2)}).} \tag{18}$$

Proof. **Step 1: Relational Invariant Form.** We start with the energy invariant relation $W = \frac{1}{2}(\kappa^2 - \beta^2)$. For a bound orbit, $W = R_s/4a$. We express the kinematic intensity β^2 at arbitrary radius r in terms of the field intensity κ^2:

$$\frac{1}{2}(\kappa^2 - \beta^2) = \frac{R_s}{4a} \quad \Longrightarrow \quad \beta^2 = \kappa^2 - \frac{R_s}{2a}. \tag{19}$$

Substituting $\kappa^2 = R_s/r$:

$$\beta^2 = R_s(\frac{1}{r} - \frac{1}{2a}). \tag{20}$$

Step 2: Constraint via Observables. We substitute the expressions for κ^2 and β^2 into the exact observable constraint $\tau_{Wo}^2 = (1-\kappa^2)(1-\beta^2)$:

$$\tau_{Wo}^2 = (1 - \frac{R_s}{r})(1 - R_s[\frac{1}{r} - \frac{1}{2a}]). \tag{21}$$

Let $A = \frac{1}{r}$ and $B = \frac{1}{r} - \frac{1}{2a}$. The equation becomes:

$$\tau_{Wo}^2 = (1 - AR_s)(1 - BR_s) = 1 - (A+B)R_s + ABR_s^2. \tag{22}$$

Rearranging into quadratic form:

$$(AB)R_s^2 - (A+B)R_s + (1 - \tau_{Wo}^2) = 0. \tag{23}$$

Step 3: Coefficient Expansion. We compute the coefficients explicitly:

$$AB = \frac{1}{r}(\frac{2a-r}{2ar}) = \frac{2a-r}{2ar^2} \tag{24}$$

$$A + B = \frac{1}{r} + \frac{2a-r}{2ar} = \frac{2a+2a-r}{2ar} = \frac{4a-r}{2ar} \tag{25}$$

Multiplying the entire quadratic equation by $2ar^2$ to clear denominators:

$$(2a-r)R_s^2 - r(4a-r)R_s + 2ar^2(1 - \tau_{Wo}^2) = 0. \tag{26}$$

Step 4: Solution. Solving for R_s:

$$R_s = \frac{r(4a-r) \pm \sqrt{r^2(4a-r)^2 - 4(2a-r)(2ar^2)(1-\tau_{Wo}^2)}}{2(2a-r)}. \tag{27}$$

Factoring out r^2 from the radical term $\sqrt{r^2(...)} = r\sqrt{(...)}$:

$$R_s = \frac{r}{2(2a-r)}((4a-r) \pm \sqrt{(4a-r)^2 - 8a(2a-r)(1-\tau_{Wo}^2)}). \tag{28}$$

For stable orbits, we select the negative root, yielding the exact algebraic link between the observed light phase and the system's geometric scale. □

1.3 The Role of G as Translation Constant

The presence of R_s in these formulas does not imply a dependency on mass m_0 or the constant G. The objection that $R_s = 2Gm_0/c^2$ makes G fundamental rests on a categorical error: it mistakes a unit conversion factor for a physical source.

Theorem 1.6 (Constants as Converters)**.** *In WILL RG, G and m_0 are derived calibration tools used to translate geometric scales into legacy units.*

Proof. The operational procedure is strictly geometric:

1. **Measure:** Light phase τ_W (dimensionless) via spectroscopy.
2. **Measure:** Geometric scale r (meters/AU) via astrometry.
3. **Calculate:** System Scale $R_s = f(r, \tau_W)$ via Theorems above.

The physical calculation ends here. The system is fully defined. If, and only if, one wishes to interface with legacy catalogues, one employs the **unit converter**:

$$m_0 \equiv \frac{R_s c^2}{2G}. \tag{29}$$

The constant G describes the units we use (kilograms vs. meters), not the physics of the system. □

Remark 1.7 (Historical Artifact)**.** *The kilogram is a human convention. In WILL RG, the fundamental quantity is the dimensionless ratio $\kappa^2 = R_s/r$, which encodes the energy density ratio $\rho/\rho_{\max}$. The "mass" m_0 is a secondary bookkeeping device.*

1.4 Summary

We have derived three algebraically distinct formulas for R_s, each optimized for different observational scenarios:

1. **Two-point** Requires two velocity measurements
2. **Balance point** Simplified form at special phase
3. **Arbitrary phase** Works for any single epoch

All formulas are:

- **Operationally independent** of G and m_0
- **Non-circular** (inputs are direct observables)
- **Algebraically exact** (no approximations beyond Keplerian closure)

The operational input is always $\tau_W = 1/[(1 + z_\kappa)(1 + z_\beta)]$, a dimensionless quantity directly measurable from spectroscopy. This demonstrates that WILL RG formulas are empirically grounded and do not rely on hidden assumptions about gravitational constants.

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